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NAME: Sohail Ansari STD.: _____ SEC.: _____ ROLL NO.: _____ SUB.: ~~English~~ ~~Maths~~

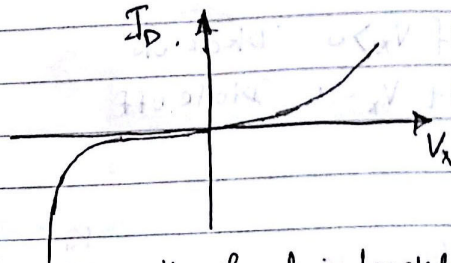
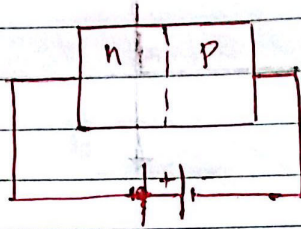
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LECTURE - 07

classmate

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REVERSE BREAKDOWN



→ In this case with $\uparrow$ in -ve voltage the junction breakdown & move in to ~~break~~ reverse breakdown region where very large current flows.

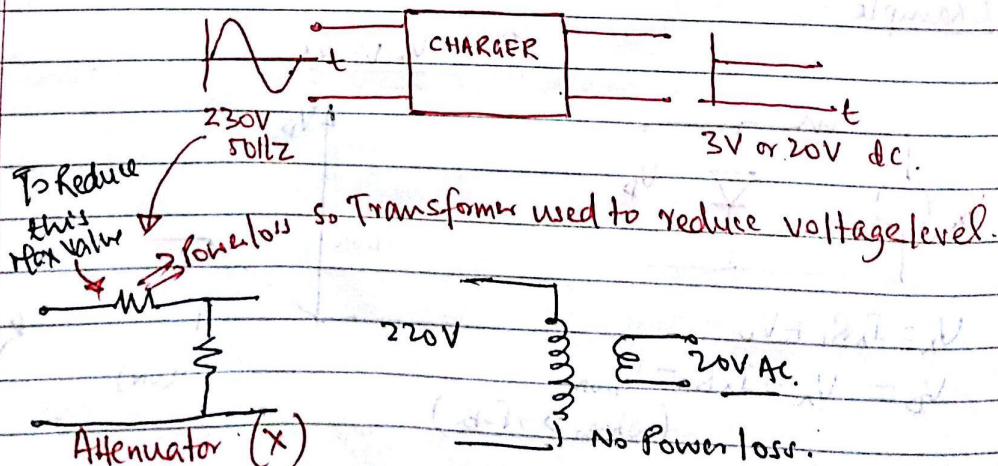
Lecture : 07

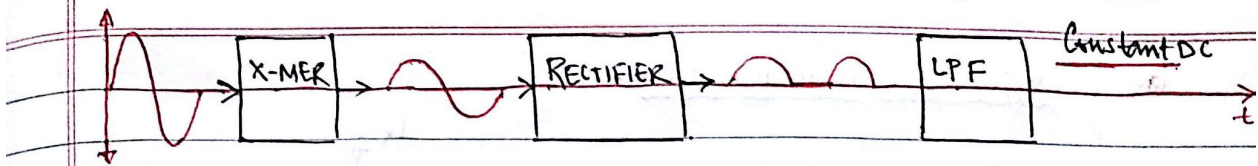
Properties:

Diode Models : exp, Constant voltage, Ideal

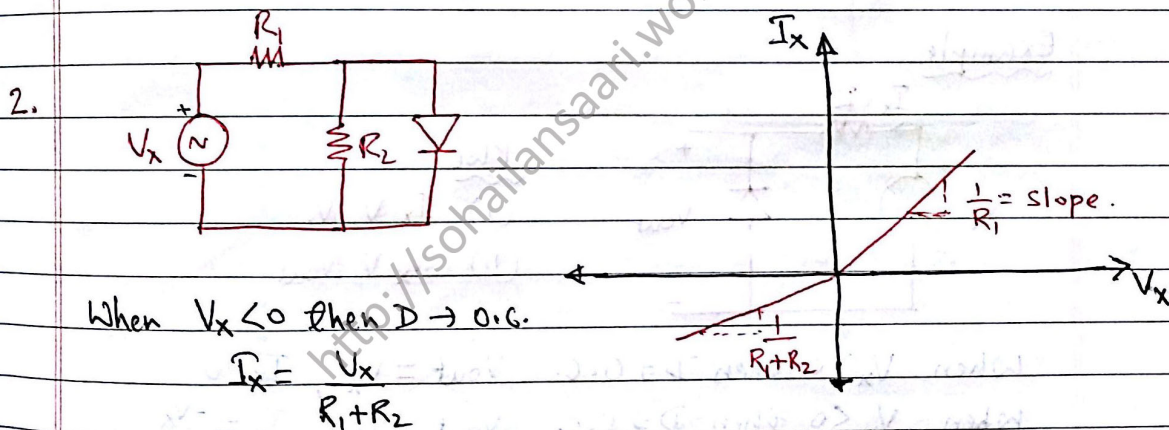
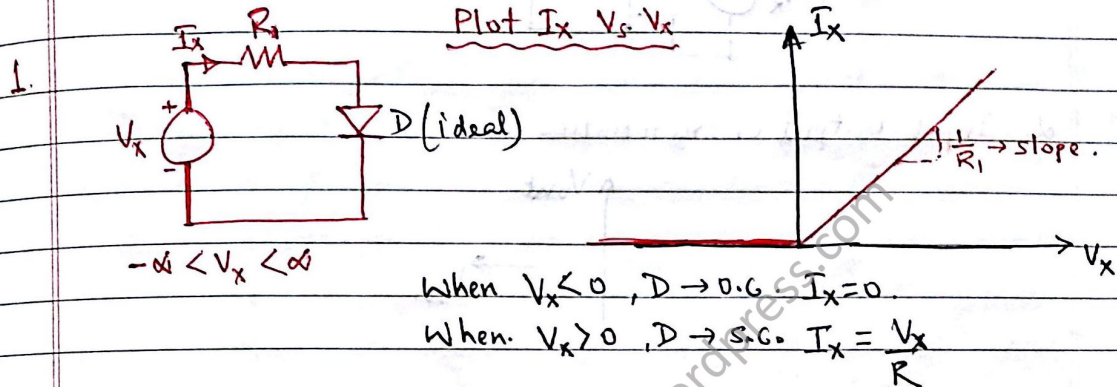
- All three model are Non-linear.
- All three model distinguish b/w +ve & -ve voltage (Rectification)
- The ideal model passes +ve voltage. The constant voltage model passes voltages greater than V_{DON} . Exponential model passes voltage if $V_D > 700-800\text{mV}$.

Application Example : charger / Adapter



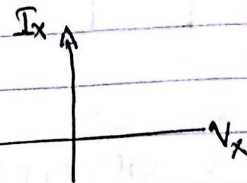
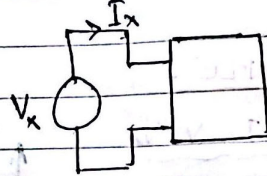


DIODE CIRCUITS EXAMPLE :

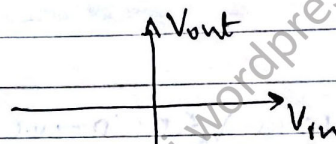


* Types of characteristics:

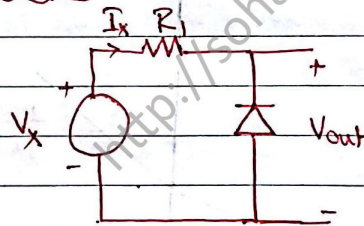
1. I-V characteristic



2. Input-output characteristics



Example

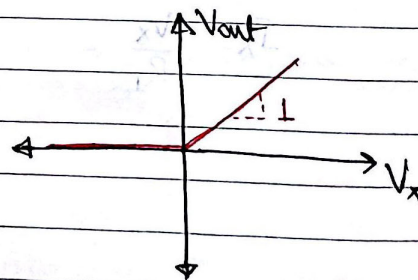
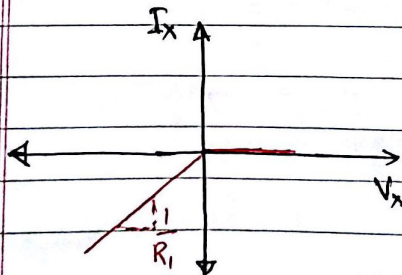


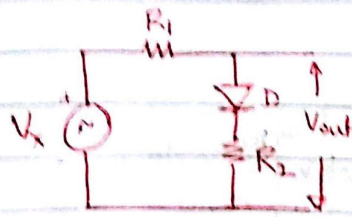
Plot

- (i) I_x, V_x, V_x
- (ii) V_x, V_s, V_{out}

When $V_x > 0$ then $D \Rightarrow$ O.C. $V_{out} = V_x, I_x = 0$

When $V_x < 0$ then $D \Rightarrow$ S.C. $V_{out} = 0, I_x = \frac{-V_x}{R_1}$



Example

Plot

(i) I_x, V_s, V_x

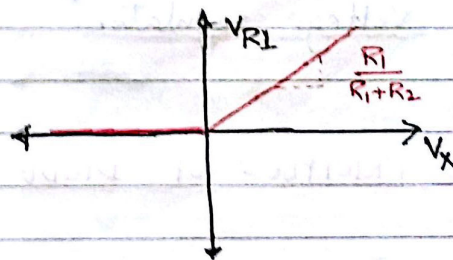
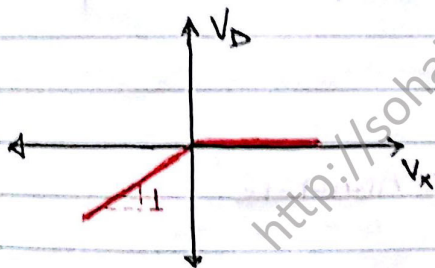
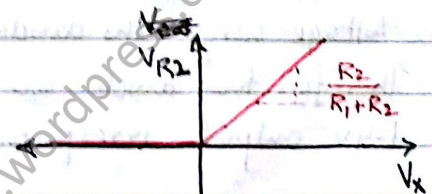
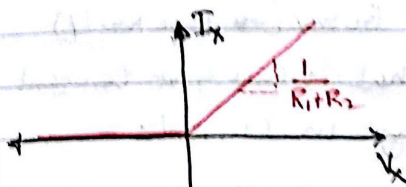
(ii) V_{R2}, V_s, V_x

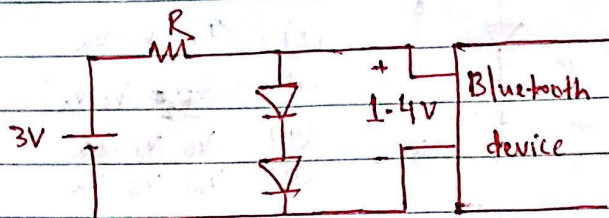
(iii) V_D, V_s, V_x

(iv) V_{R1}, V_s, V_x

When $V_x < 0$ $D \rightarrow o.c.$, $I_x = 0$, $V_{out} = V_x$, $V_{R2} = 0$, $V_D = V_x$

When $V_x > 0$ $D \rightarrow s.c.$, $I_x = \frac{V_x}{R_1 + R_2}$, $V_{out} = I_x R_2 = \frac{V_x R_2}{R_1 + R_2}$, $V_D = 0$



Application Example

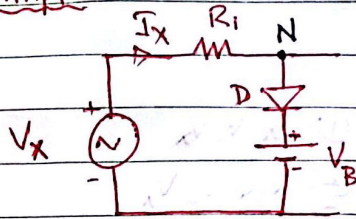
Why two diode are used why not Resistor?

- The voltage across the diode is a weak sensation of Current. ~~So small~~ For 10 times ↑ in Current the Voltage across the diode is only 60mV (see Page No: 17). Therefore two diode are used so that it can give constant 1.4V output irrespective of change in Current.

- Hence we can say that Diode operate as Simple Voltage regulator.

- PRICIPLES OF DIODE CIRCUIT ANALYSIS

- Begin by Assuming certain states for all diodes.
- check the Final results against the Assumption.
- If a diode is about to turn on or off, it must sustain a voltage of $V_{D(on)}$ but its Current is small.
- If a diode is ON & carries a Current, the Current must flow from Anode to Cathode.

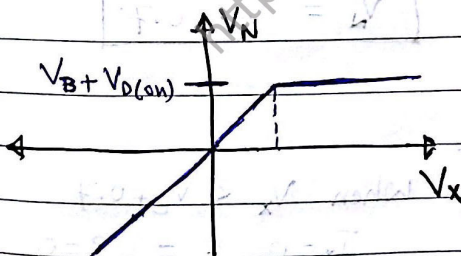
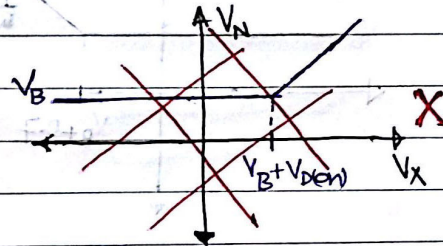
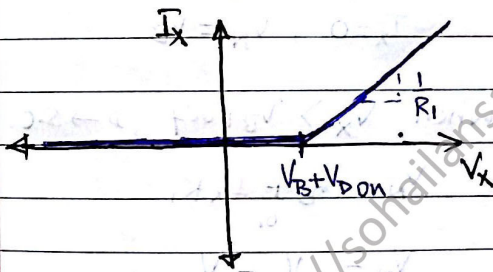
ExamplePlot

- ① I_x vs V_x
- ② V_N vs V_x

When $V_x \geq V_B + V_{D(on)}$ then $D \rightarrow$ Short circuit $\rightarrow$ why?

When $V_N = V_B + V_{D(on)} \rightarrow D$ begins to turn on. and at this point current is very small (see principle of Diode Ckt (Pt No-3)). So, voltage across R_i is small. Hence value of V_x is approximately equal to V_N . Hence, $I_x = \frac{(V_x - V_B - V_{D(on)})}{R_i} = \frac{(V_x - V_B - V_{D(on)})}{R_i}$

When $V_x < V_B + V_{D(on)}$ then $D \rightarrow$ open circuit, $I_x = 0$.



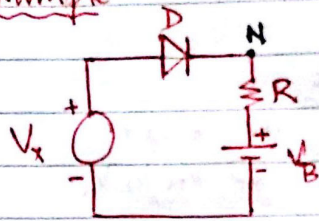
When $V_x < (V_B + V_{D(on)})$
 $D \rightarrow$ o.c.

$$V_N = V_x$$

When $V_x > (V_B + V_{D(on)})$
 $D \rightarrow$ s.c.

$$V_N = V_B + V_{D(on)}$$

Example

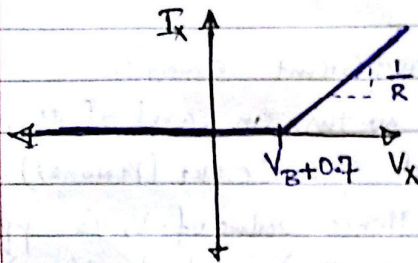


Plot

① I_X vs V_X

② V_N vs V_X

③ V_R vs V_X

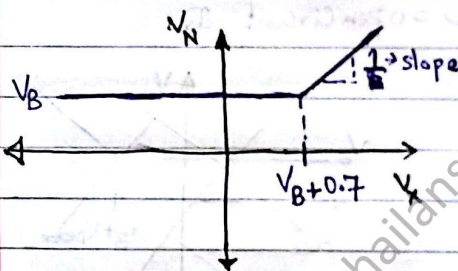


When $V_X < V_B + 0.7$ $D \rightarrow o.c.$

$$I_X = 0$$

When $V_X > V_B + 0.7$ $D \rightarrow s.c.$

$$I_X = \frac{V_X - V_B - 0.7}{R}$$



When $V_X < V_B + 0.7$, $D \rightarrow o.c.$

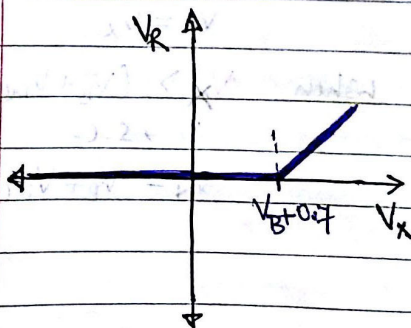
$$I_X = 0, V_N = V_B$$

When $V_X > V_B + 0.7$, $D \rightarrow s.c.$

$$V_N = V_B + I_X R$$

$$V_N = V_X - V_{D(\text{on})}$$

$$V_N = V_X - 0.7$$



When $V_X < V_B + 0.7$

$$I_X = 0, V_R = I_X R = 0$$

When $V_X > V_B + 0.7$

$$V_R = V_X - 0.7 - V_B$$